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Deep Learning Beehive Monitoring System for Early Detection of the Varroa Mite

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Abstract: One of the most critical causes of colony collapse disorder in beekeeping is caused by the Varroa mite. This paper presents an embedded camera module supported by a deep learning algorithm for the process of early detecting of Varroa infestations. This is achieved using a deep learning algorithm that tries to identify bees inside the brood frames carrying the mite in real-time. The end-node device camera module is placed inside the brood box. It is equipped with offline detection in remote areas of limited network coverage or online imagery data transmission and mite detection over the cloud. The proposed deep learning algorithm uses a deep learning network for bee object detection and an image processing step to identify the mite on the previously detected objects. Finally, the authors present their proof of concept experimentation of their approach that can offer a total bee and varroa detection accuracy of close to 70%. The authors present in detail and discuss their experimental results.

Keywords: beehive monitoring systems; Varroa mite detection; Internet of Things (IoT); Convolutional Neural Networks (CNNs); image processing; performance evaluation; distributed systems

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1. Introduction

Bees' population declination worldwide is mainly caused due to the reckless use of pesticides by humans [1], as well as the emergence of new resistant strains of bacteria and mites. Varroa disease is one of the most resistant diseases with two main forms (Varroa destructor and Varroa jacobsoni) [2], affecting the growth of the offspring, stressing the bees, and reducing their chances of survival in winter. More specifically, Varroa mite is one of the most devastating diseases for the beekeeping population. This parasite attaches to the bee and feeds on its fat cells, causing wing deformations.

The environment inside and around the beehives is vital to the colony establishment's success and development. An essential factor in apiary hives that affects both colony survival and honey yield is the ability to manage agricultural interventions and disease treatments (especially Varroa mite [3]) and monitor the conditions inside the beehive [4–6].

Most beekeepers use synthetic pesticides to control the mite infestation; nevertheless, the mite soon develops resistance to their active compounds, compromising their effectiveness. The prevalence of the mite over the year is confronted by: (a) The reproduction of resistant Apis mellifera breeds [7] and (b) the appliance of precise interventions upon the detection of the mite that includes controlled high temperatures (above 38 °C) or increased relative humidity stress events above 70% for normal brood box temperature environments (32–37 °C) [8,9].

The Internet of Everything (IoE) industry is shifting fast towards the beekeeping sector, aiming for the vast applicability of Internet technologies and IoT, capable of applying new smart AI detection algorithms [10–12]. Existing applications in agriculture include environmental monitoring and bee monitoring IoT systems [13–18]. This paper focuses on technological solutions for detecting bees and then Varroa mites using cameras inside the brood box.

- The real-time clock module. An Inter-Integrated Circuit (I2C) DS1307 Real Time Clock (RTC) module is placed at the General Purpose Input Output (GPIO) pin of the MCU for the process of keeping track of time if the end-node device is operating in offline mode. Suppose online appropriate Network Time Protocol (NTP) service is automatically used to calibrate time offsets and reset the RTC DateTime.
- The Wi-Fi concentrator. It is used only in the online mode of operation, and it is a Wi-Fi access point device that includes an LTE/3G cellular transceiver. The end-node MCU connects to the concentrator for the process of imagery data uploads to the cloud if the device operates in online mode. For the offline mode, the MCU BLE interface is used, transmitting to a distance up to 4–10 m the detection output of the beehive. The following subsection describes the end-node device's two modes of operation.

Figure 1. Proposed Varroa mite detection system end-node device and device components.

3.2. End-Node Device Functionality and Modes of Operation

There are two modes of operation that the end-node device can be used: Offline and online. The operation modes can be selected using a selection switch placed at the bottom of the MCU casing. The switch is connected to an MCU GPIO pin digital input, and by setting it to LOW or HIGH, it switches between the two modes of operation accordingly. The two end-node modes operate as follows:

- **Online mode:** In this mode of operation, the process of Varroa detection is performed over the cloud. For this purpose, appropriate cloud service and API using HTTP requests have been implemented. The API is capable of image data transmission from the uploaded by the end-node MCU using HTTP protocol PUT requests. Also, an HTTP JSON POST request can be sent to the cloud API, including an API key and a beehive id, and the API returns as part of a JSON object. The Varroa detection results for this beehive, including the base64 encoded images Regions Of Interest (ROIs), where Varroa mite has been detected.
  The online mode of operation requires using the beehive concentrator, which is responsible for the node data transmissions over the internet over HTTP. It acts as an intermediate gateway among the end nodes and the cloud application service. The concentrator can upload images with an overall bandwidth capability that varies from 1–7/10–57 Mbps, depending on the gateway distance from the beehive [39]. If the distance is 20–30 m, it is limited by the LTE technology used.

- **Offline mode:** In places of limited Internet connectivity and cellular coverage, the end-node offline mode can be used. The offline mode does not require the use of the concentrator device.
  In this mode, a micro-service that includes a version of the detection algorithm inside the end-node device is used for the process of executing the Varroa detection algorithm locally (as presented in Figure 2). Then, the final CSV output is transmitted using the MCU BLE transponder to the farmers' mobile phones. A BLE service and two read characteristics can be used for the CSV output, and Varroa detected ROI image acquisition accordingly. The beekeeper can check the status of each one of his beehives by moving close to the beehive and pairing with each one accordingly, performing the BLE read from his mobile phone. The drawbacks of the offline mode are that it offers 20–25% less end-node energy consumption and no communication provider costs. Nevertheless, it has difficulties with BLE pairing, especially if many BLE devices are close-by and there are difficulties on characteristic reads of imagery base64 encoded data [40,41]. For this reason, only one Varroa mite ROI is available (the last one detected) via that BLE characteristic.

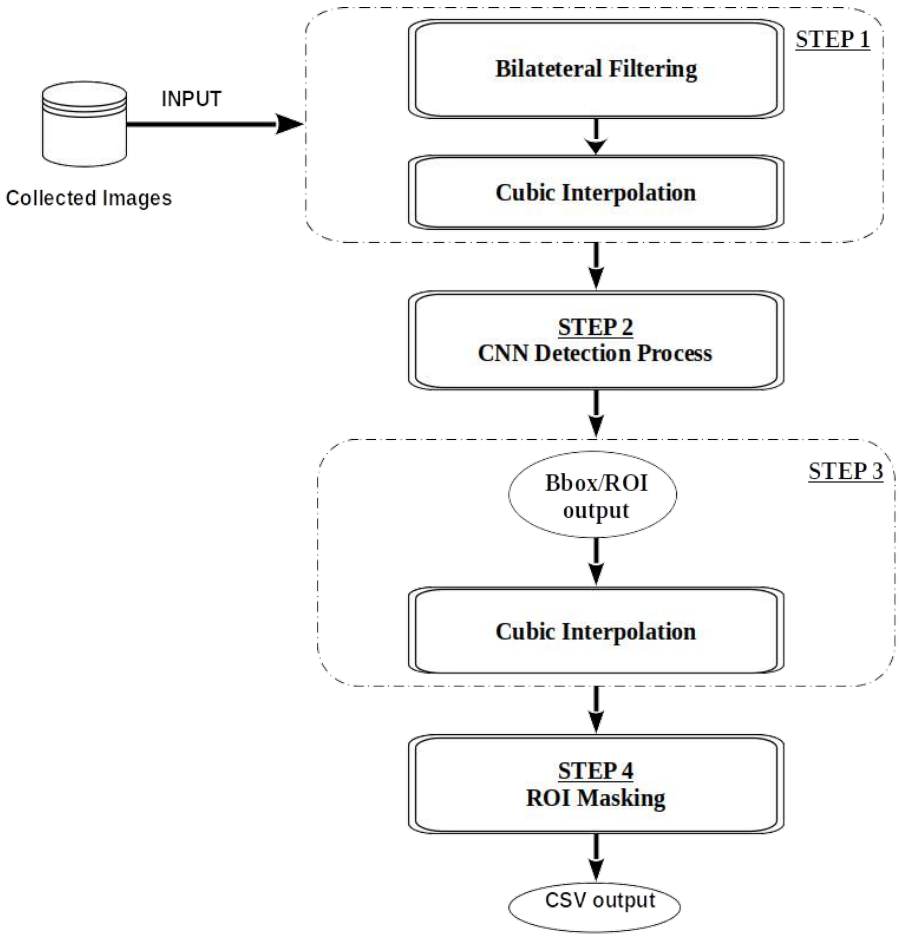

Figure 2. Proposed algorithm process flowchart for the detection of Varroa mite.

The following section describes the Varroa mite detection algorithm used by the end-node devices in offline or online operation modes.

4. Proposed Method for Varroa Mite Early Detection

Training a Convolutional Neural Network for an object initiates with the collection of the sample of images that contain the objects to be detected. If no object localization applies to the image, then the process involved is a classification process that assists in classifying

applied to detect circular areas with a lower-upper threshold of 10–90 px^2. If at least one circular area of this threshold is detected on a bee object, this bee is set as detected with Varroosis. The original color image, including the detected bee object contours with the masked Varroa mite areas annotated on each CNN detected image, is stored in the appropriate output folder. The detection results are appended to the detection service process CSV output file.

- **Detection service process and data output:** The detection process is performed by a daemon application that is installed as a service on a cloud server or at the embedded end-node device depending on the mode of operation (online, offline). This application loads the inference graph of the CNN neural network into the system memory so that the bees can be detected and then the Varroa mite can be detected. This procedure is performed on each image received from the end node device using HTTP PUT requests. The HTTP PUT method requires that the requested URI message be updated or created, which is enclosed in the body of the PUT message. Thus, if there is a resource in this URI, the message body is considered as a new modified version of this resource. Once the PUT request is received, the service starts scanning the bees and then the Varroa so you can output an updated CSV file containing the number of Varroa mites detected in each photo taken by the end node device. Figure 2 shows in detail the steps of the detection process implemented on the cloud server (online) or in the embedded end-node device (offline).

The following section presents the authors' experimental results using the proposed CNNs combined with computational vision techniques. Figure 3 shows the detection output from each one of the steps that are followed by the author's proposed detection process as illustrated in the flowchart in Figure 2. The images displayed/per detection step have been taken during the Varroa mite detection process validation.

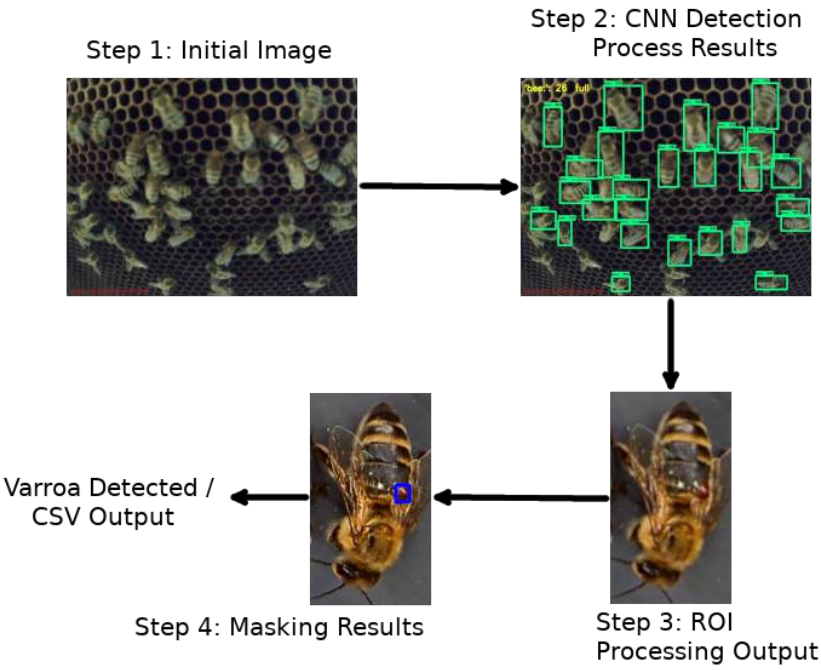

Figure 3. Proposed algorithm step-by-step detection output per flowchart detection process step.

5. Experimental Scenarios and Results

This section presents the experimental scenarios that were implemented using end-node devices to detect bees using a selected CNN (Faster R-CNN), that uses the region proposal network (PRN) to reduce the total image detection times at least 10× times [44], and one of three pre-trained models: MobileNet V2 [55], MobileNet V3 [56], or ResNet-50